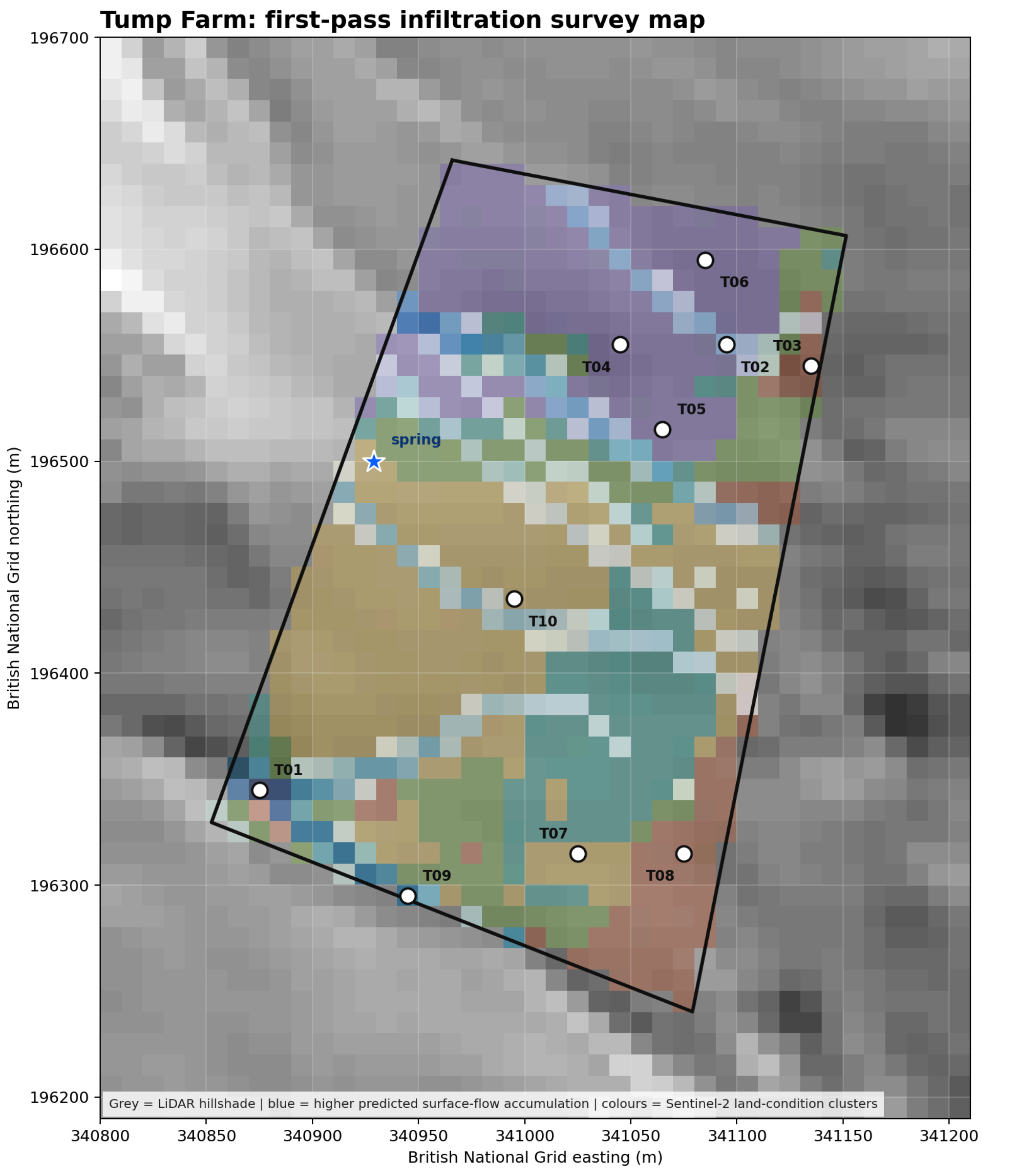
Tump Farm Remote-Sensing Infiltration Survey Plan

Purpose: choose a small number of infiltration-test locations that compare different land-condition signals and different hydrological positions before the site visit.



Numbered Test Locations

ID	Coordinate	Satellite zone	Reason for testing
T01	51.662760, -2.856217	weaker / drier vegetation	High predicted convergence with weaker/drier vegetation.

ID	Coordinate	Satellite zone	Reason for testing
T02	51.664671, -2.853073	vigorous / moist vegetation	High predicted convergence with vigorous/moist vegetation.
T03	51.664586, -2.852493	weaker / drier vegetation	Low-flow hillslope with weaker/drier vegetation.
T04	51.664666, -2.853795	vigorous / moist vegetation	Low-flow hillslope with vigorous/moist vegetation.
T05	51.664309, -2.853499	vigorous / moist vegetation	Persistent moisture signal; investigate whether it is soil, vegetation or drainage.
T06	51.665030, -2.853224	vigorous / moist vegetation	Seasonally variable vegetation; useful check on management or moisture sensitivity.
T07	51.662506, -2.854044	mixed pasture condition	Paired comparison across an apparent land-condition boundary, side A; placed into the zone core where possible.
T08	51.662512, -2.853321	weaker / drier vegetation	Paired comparison across an apparent land-condition boundary, side B; placed into the zone core where possible.
T09	51.662318, -2.855197	mixed pasture condition	Highest local flow accumulation away from the spring exclusion zone.
T10	51.663582, -2.854498	mixed pasture condition	Representative point for mixed pasture condition.

Method

The terrain layer uses the existing 2 m LiDAR-derived flow accumulation for the spring area. The vegetation layer uses Sentinel-2 Level-2A scenes from 2024-01-16 (7.4% cloud), 2024-06-02 (2.9% cloud), 2024-11-26 (0.1% cloud), 2025-04-02 (0.3% cloud). For each clear scene, NDVI and NDMI were calculated inside the assumed workable boundary. Pixels were clustered by median vegetation vigour, median moisture signal and seasonal variability.

The test points were then selected by simple stratified rules: high-flow versus low-flow topographic positions, stronger versus weaker vegetation condition, persistent moisture signals, seasonal variability, and paired points across the sharpest apparent land-condition boundary.

Land-Condition Clusters

Cluster	Interpretation	NDVI median	NDMI median	NDVI variability
0	weaker / drier vegetation	0.602	0.102	0.211
1	mixed pasture condition	0.686	0.178	0.232
2	mixed pasture condition	0.677	0.241	0.196
3	mixed pasture condition	0.749	0.291	0.225
4	vigorous / moist vegetation	0.774	0.352	0.265

How To Use This In The Field

At each numbered point, run the same infiltration-ring protocol, take a photo, record ground cover, visible compaction, grazing pressure, slope position, current moisture, drains/pipes/culverts, and farmer comments. For boundary-paired tests, avoid the fence line itself: place the ring comfortably inside the apparent zone, ideally 10-20 m from the physical boundary if the field allows it.

The ring test is mainly measuring near-surface infiltration behaviour and saturated hydraulic conductivity at a small patch scale. That is useful for comparing trampling, grazing compaction, surface crusting and root/macropore effects, but it is not a direct spring-recharge measurement.

> **Limit:** this map identifies contrasts worth checking. It does not prove why those contrasts exist, and it does not prove spring recharge. Sentinel-2 is 10 m scale for the main vegetation bands, so narrow hedges and fence-line changes can be mixed across pixels. NDMI is also weather-sensitive, especially immediately after rain. A DEM routes surface runoff if runoff occurs; infiltration tests help estimate whether runoff is actually generated.